# Changqi Sun

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## SUMMARY

Fourth-year Computer Science PhD student at Virginia Tech (expected Jun 2028) in applied cryptography, with a **BS in Mathematics**. Implements and hardens NIST post-quantum signatures. **Side channels:** hardened a **masked ML-DSA (Dilithium, FIPS 204)** signer on **Arm Cortex-M4**, showing register-level countermeasures leak on buses and ports and under **-O2** (TVLA, CPA and template attacks in simulation), and cut peak signing RAM by **37–42%**. **Threshold PQC:** co-first author and presenter of a full-threshold ML-DSA framework whose signatures are identical to standard FIPS 204 signatures (**IEEE S&P 2026**); first author of FLOCK, a **FALCON-compatible threshold signature** scheme (under submission at **USENIX Security 2027**). **Open-source PQC code:** root-caused an under-seeded discrete-Gaussian sampler in an ACM CCS 2026 threshold-Falcon artifact (**patch merged upstream**, ∼**9**× faster signing). C/C++ (primary), Rust, Python. Native Mandarin speaker. Seeking a full-time **Summer 2027** internship in **Beijing**.

## EDUCATION

**Virginia Tech** , *PhD. in Computer Science & Application* | Blacksburg , US GPA: 3.64 / 4.0 — Aug 2023 - Jun 2028 (expected)

**Shandong University**, *MA. in Finance* | Shandong , China GPA: 83.41 / 100 — Sep 2017 - Jun 2019

**Northeast Forestry University**, *BS. in Mathematics* | Heilongjiang , China GPA: 81.88 / 100 — Sep 2012 - Jun 2016

## PUBLICATION

Sedghighadikolaei, Kiarash*, **Sun, Changqi***, Hoang, Thang, Hamdaoui, Bechir, and Yavuz, Attila A. "A Full Threshold NIST PQC-Compliant Framework for Distributed Trust in Federal Public Key Infrastructure." *IEEE Symposium on Security and Privacy (S&P 2026)*, Accepted. *Equal contribution; presenter.

**Sun, Changqi**, Sedghighadikolaei, Kiarash, Yavuz, Attila A., Mozaffari Kermani, Mehran, Hamdaoui, Bechir, and Hoang, Thang. "FLOCK: Efficient FALCON-Compatible Threshold Signatures." Under submission at *USENIX Security 2027*.

**Sun, Changqi**, Nguyen, Hoang-Dung, Chandler, Jack, Xiong, Wenjie, and Hoang, Thang. "HERO: Fast and Simple ORAM with a Little Friend." Under review at *NDSS 2027*.

**Sun, Changqi**, and Li, Rui. "RC5-CBC: Best Matching Combination of Block Cipher and Model of Operation for Implementing ORAM Schemes." *International Conference on Cryptography, Network Security, and Communication Technology (CNSCT 2023)*, Proceedings of SPIE, Vol. 12641, 2023.

## PROJECTS

**Side-Channel-Hardened, Memory-Reduced Masked ML-DSA** | *C, Cortex-M4, TVLA* — Sep 2026 – Present

- Hardened a **masked ML-DSA (FIPS 204)** signer in C (built on `mldsa-native`, incl. masked Keccak) for **Arm Cortex-M4**. Showed that register-focused countermeasures fail on the compiled binary: shares meet on buses and ports ($|t| = 199$ under a Hamming-distance bus model while the register model is clean), and a register-clearing barrier holds at -O3 but leaks at -O2.
- Fixed both with **bus-and-port barriers**, three Boolean check shares, and **full-register inline-assembly barriers**: no detections remain in the affected regions under the bus and operand models.
- Cut peak signing RAM by **37.1–41.9%** across ML-DSA-44/65/87 (ML-DSA-44: **50,612** → **31,828 B**, under 32 KiB) via public-data packing, reuse of dead share buffers, and seed-derived key shares, for only **+0.21%** instructions.
- Built the evaluation in **Cortex-M4 emulation**: first/second-order **TVLA** under register, bus, and operand leakage models over full signing (up to **267.5M instructions per trace**), CPA and template attacks, and binary-level **constant-time** checking, each with positive controls. Manuscript in preparation for **IEEE S&P 2027**.

**Upstream Fix in the ACM CCS 2026 Threshold-Falcon Artifact** | *C++* | Merged PR — 2026

- Diagnosed signatures failing the `l2bound` check on *consecutive* retries while reproducing the artifact for *Thresholdizing Standardized FALCON Signatures* (ACM CCS 2026) — statistically impossible at Falcon's design rejection rate, which localized the fault to the sampler rather than to chance.
- Root-caused an **under-seeded discrete-Gaussian sampler**: a byte/element unit mismatch against `emp-tool`'s `PRG::random_data` left **7/8 of the sampler buffer read from uninitialized heap memory**, skewing signature norms ≈**15%** toward the rejection bound.
- Fixed a second defect where `read()`/`write()` returns stored in an unsigned `size_t` turned an error code of −1 into `SIZE_MAX`, silently desynchronizing the I/O stream.
- Validated with A/B measurements: mean squared signature norm within **0.4%** of its expected value and **zero** rejections, removing the repeated retries that made affected runs ∼**9**× slower. **Pull request merged by the paper's authors.**

**Full-Threshold ML-DSA via Malicious Mixed-Mode MPC** | *C++, S&P 2026* | GitHub Aug 2024 – Aug 2025

- Designed a **full-threshold ($n$-of-$n$)** signing framework for **ML-DSA (NIST FIPS 204)**: after a one-time trusted setup, $n$ signing servers jointly run the unmodified FIPS 204 signing algorithm inside MPC, so no single server holds a usable key and every signature is **identical to a standard one** (verifies under the unmodified verifier). Co-first author and presenter.
- Security **with abort against up to** $n-1$ **malicious servers** in the **quantum-UC** model, combining SPDZ for arithmetic steps with WRK17 authenticated garbling for bit-level steps (SHAKE/Keccak-f, Decompose, SampleInBall, CheckNorm, MakeHint), bridged by daBits.
- Synthesized the bit-level steps from Verilog with Yosys into Bristol-format circuits (SampleInBall **890K → 355K gates** vs. a SCALE-MAMBA baseline); benchmarked 2–5 servers across ML-DSA-44/65/87.

**Asymmetric Distributed ORAM** | *C/C++, NTL, OpenSSL, Access-Pattern Privacy* | GitHub Aug 2023 – Present

- Encryption hides what records contain but not *which* records are accessed. Designed an **asymmetric distributed ORAM** in which an untrusted, non-colluding helper assists eviction with resource usage **independent of data size**. First author; under review at **NDSS 2027**.
- For an earlier trusted-helper variant (not part of the NDSS design), ported the prototypes to **Intel SGX**: worked through linking **NTL** into the enclave, moved to an **OpenSSL**-based path, and investigated a cross-server slowdown with multithreading microbenchmarks.
- Modularized **ElGamal** behind interchangeable implementation interfaces and shipped a documented software prototype to collaborators working on **FPGA-based SmartNIC** integration.
- Benchmarked client–server–helper deployments on Windows, Linux, and a Raspberry Pi 5 helper, with scripted local/remote builds, runs, and log collection across block-size and tree-height sweeps.

## PROFESSIONAL SERVICE

Program Committee Member, Artifact Evaluation Committee, ACM Conference on Computer and Communications Security (ACM CCS) 2026

## RESEARCH EXPERIENCE

**Virginia Tech,** *Graduate Research Assistant* | In person (Blacksburg, Virginia) Aug. 2023 – Present

- Conduct research on **secure distributed systems**, including threshold post-quantum signatures, distributed trust via MPC, and Oblivious RAM (ORAM).
- Designed and implemented cryptographic and systems prototypes in **C/C++**, with emphasis on networking, benchmarking, and hardware/software integration.
- Serve on the **ACM CCS 2026 Artifact Evaluation Committee**; maintain reproducible build, benchmark, and experiment documentation so research prototypes survive handoff.

## TEACHING EXPERIENCE

**Virginia Tech,** *Graduate Teaching Assistant* | Blacksburg, Virginia Aug. 2023 – Present

- CS 5740 (AI Tools for Software Engineering), Fall 2026 — graduate course on **applied generative AI for software developers**: LLM-based code generation and comprehension, automated testing and documentation, and evaluation of AI coding tools.
- CS 5134 (Introduction to Quantum Computer Science), Fall 2025 — developed lecture notes and course materials on universal gate sets, **Shor's and Grover's algorithms**, noise and **quantum error correction**, and quantum Shannon theory (entropy, entanglement measures, teleportation, superdense coding).
- CS 5644 (Machine Learning with Big Data), Spring 2025 — created programming assignments on SVMs, random forests, and transformer fine-tuning; held office hours and graded semester-long ML projects.
- CS 4104 (Data & Algorithm Analysis), Fall 2023 – Fall 2024 — designed homework problems on greedy and dynamic-programming algorithms.

## SKILLS

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| **Embedded Crypto & SCA:** | Arm Cortex-M4 (GCC inline-assembly barriers, compiler-level leakage at -O2/-O3), Boolean/arithmetic masking, masked Keccak / SHA-3, TVLA (1st/2nd order), CPA and template attacks, Hamming-weight/distance leakage models, constant-time checking, Unicorn emulation |
| **Cryptography & Security:** | ML-DSA (FIPS 204), Falcon, threshold signatures, secure multi-party computation (malicious security), Oblivious RAM, Intel SGX / TEE, UC security proofs |
| **Programming:** | C/C++ (primary), Python, Rust, TypeScript/JavaScript, SQL, R |
| **Mathematics:** | Lattice-based cryptography, discrete-Gaussian sampling, probability and statistics, linear and nonlinear optimization; BS in Mathematics |